1 **(a)** 9 **[B1]**

(b) 2 **[B1]**

(c) Isosceles triangle **[B1]**

[Total Marks: 3]

2 **(a)** 10 **[B1]**

(b) Rhombus OR Rectangle **[B1]**

(c) 72 **[B1]**

[Total Marks: 3]

3 **(a)** **(i)** 6 **[B1]**

(ii) 6 **[B1]**

(b) **(i)** 1 **[B1]**

(ii) 1 **[B1]**

[Total Marks: 4]

4 **(a)** 4 **[B1]**

(b) 0 **[B1]**

(c) Correct reflection of at least two 'petals' (e.g. vertices at $(1, 4)$ and $(4, 3)$) **[M1]**

Fully correct shape drawn **[A1]**

[Total Marks: 4]

5 **(a)** **(i)** 15 **[B1]**

(ii) 15 **[B1]**

(b) Rectangle OR Rhombus **[B1]**

(c) 5 **[B1]**

[Total Marks: 4]

6 Parallelogram: 0 lines, order 2 **[B2, B1 for each correct value]**

Regular Pentagon: 5 lines, order 5 **[B2, B1 for each correct value]**

Isosceles Trapezium: 1 line, order 1 **[B2, B1 for each correct value]**

[Total Marks: 6]

7 **(a)** **(i)** 0 **[B1]**

(ii) 2 **[B1]**

(b) Both squares in the left column row 2 and right column row 3 shaded (or top row column 2 and bottom row column 3 shaded) **[B2]**

(c) Lines of symmetry = 12 **[B1]**

Exterior angle = 30° **[B1]**

[Total Marks: 6]

8 **(a)** Identifies 6 vertical planes of symmetry or 1 horizontal plane of symmetry **[M1]**

7 **[A1]**

(b) 6 **[B1]**

(c) 4 **[B1]**

[Total Marks: 4]

9 **(a)** **(i)** 6 **[B1]**

(ii) 4 **[B1]**

(iii) Infinite **[B1]**

(b) Exterior angle $= \frac{360}{12} = 30°$ **[M1]**

Interior angle $= 180 - 30 = 150$ **[A1]**

[Total Marks: 5]

10 **(a)** 4 **[B1]**

(b) 6 **[B2]**

Strictly 5 (vertical) or 1 (horizontal) planes identified **[B1]**

(c) A straight line passing through the apex (top vertex) **[B1]**

and the centre of the circular base. **[B1]**

[Total Marks: 5]

11 **(a)** **(i)** 6 longitudinal planes and 1 perpendicular bisecting plane seen or implied **[M1]**

7 **[A1]**

(ii) 6 **[B1]**

(b) Planes of symmetry = 5 **[B1]**

Order of rotational symmetry = 5 **[B1]**

(c) Planes of symmetry = infinite (or ∞) **[B1]**

Slant height $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Total surface area $= \pi(6)(10) + \pi(6)^2 = 96\pi$ **[A1]**

[Total Marks: 8]

12 (a) Recognises that order 4 symmetry implies point symmetry (order 2) about M, so M is the midpoint of AC **[M1]**

$(5, -2)$ **[A1]**

(b) Gradient of $AC = \dfrac{-2-6}{5-1}$ or $\dfrac{-8}{4} = -2$ **[M1]**

Substitute $(1, 6)$ or $(5, -2)$ or $(3, 2)$ into $y = -2x + c$ OR $y - 6 = -2(x - 1)$ **[M1]**

$-2x + 8$ **[A1]**

(c) Deduces that $ABCD$ must be a square because it has order 4 rotational symmetry (area is $|AB|^2$) **[M1]**

$|AB|^2 = (7-1)^2 + (4-6)^2$ **[M1]**

40 **[A1]**

[Total Marks: 8]

13 (a) 4 **[B2]**

If 0 scored, allow [B1] for identifying either 3 (vertical) OR 1 (horizontal/mid-plane).

(b) 3 **[B1]**

(c) 7 **[B2]**

If 0 scored, allow [B1] for 6.

[Total Marks: 5]

14 (a) (i) $\dfrac{360}{15}$ **[M1]**

24 **[A1]**

(ii) 24 **[B1]**

(b) Kite OR Isosceles Trapezium (accept Isosceles Trapezoid) **[B1]**

(c) Infinite (accept an infinite number) **[B1]**

[Total Marks: 5]

15 (a) (i) 4 **[B1]**

(ii) Infinite (or infinitely many) **[B1]**

(b) Uses volume formula: $\dfrac{1}{3} \times 6^2 \times h = 48$ **[M1]**

Solves for height: $h = 4$ **[A1]**

Calculates distance from centre of base to a corner $= \sqrt{3^2+3^2} = \sqrt{18}$ (or $3\sqrt{2}$ or 4.24...) **[M1]**

Uses Pythagoras' theorem for sloping edge $l = \sqrt{4^2 + (\sqrt{18})^2}$ **[M1]**

5.83 (or 5.830...) **[A1]**

[Total Marks: 7]

16 **(a)** Understanding that diagonals of a rhombus intersect at 90°, hence gradient $m = -\frac{1}{3}$ **[M1]**

$-\frac{1}{3}x$ (accept equivalent forms) **[A1]**

(b) 2 **[B1]**

(c) Exterior angle $= 180° - 165° = 15°$ **[M1]**

$n = \frac{360}{15}$ **[M1]**

$n = 24$ **[A1]**

[Total Marks: 6]

17 **(a)** **(i)** 8 **[B1]**

(ii) 9 **[B2]**

If B0, score B1 for 8 (identifying only the vertical planes)

(b) **(i)** 12 **[B1]**

(ii) $(12-2) \times 180$ **[M1]**

1800 **[A1]**

(c) $(-2, 3)$ **[B2]**

If B0, score M1 for point plotted correctly on grid OR one coordinate correct

[Total Marks: 8]

18 **(a)** Calculates length $AB = \sqrt{(5-1)^2+(4-2)^2} = \sqrt{20}$ **[M1]**

Calculates length $BC = \sqrt{(3-5)^2+(8-4)^2} = \sqrt{20}$ **[M1]**

States $AB = BC$ therefore the triangle is isosceles **[A1]**

(b) Identifies midpoint of $AC = (\frac{1+3}{2}, \frac{2+8}{2}) = (2, 5)$ **[M1]**

Calculates gradient of line of symmetry $= \frac{4-5}{5-2} = -\frac{1}{3}$ (or uses perpendicular to AC) **[M1]**

Substitutes $(5, 4)$ or $(2, 5)$ into $y = -\frac{1}{3}x + c$ to find c **[M1]**

$-\frac{1}{3}x + \frac{17}{3}$ **[A1]**

(c) Identifies $n = 15$ and uses $\frac{(15-2) \times 180}{15}$ OR $180 - (360 \div 15)$ **[M1]**

156 **[A1]**

[Total Marks: 9]

19 **(a)** 2 **[B1]**

(b) Gradient of $BD = -\frac{1}{2}$ (using $m_1 \times m_2 = -1$) **[M1]**

$-0.5x$ or $-\frac{1}{2}x$ **[A1]**

(c) **(i)** Uses Area $= \frac{1}{2} \times d_1 \times d_2$: $\frac{1}{2} \times AC \times 4\sqrt{5} = 40$ **[M1]**

$AC = \frac{80}{4\sqrt{5}} = 4\sqrt{5}$ **[A1]**

(ii) $OA = 2\sqrt{5}$, so $k^2 + (2k)^2 = (2\sqrt{5})^2$ or $5k^2 = 20$ **[M1]**

$(2, 4)$ **[A1]**

(iii) Sets up distance equation for $B(x, -0.5x)$ with $OB = 2\sqrt{5}$: $x^2 + (-0.5x)^2 = 20$ **[M1]**

$1.25x^2 = 20$ giving $x^2 = 16$ **[M1]**

$(4, -2)$ (rejects negative x due to visual quadrant) **[A1]**

[Total Marks: 10]

20 **(a)** States that a right prism with a regular n-gon base has n vertical planes of symmetry and 1 horizontal plane of symmetry (halfway along its length) **[B1]**

Sets up the equation $n + 1 = 7$, yielding $n = 6$ **[B1]**

(b) **(i)** Recognises the side length of the hexagon $= 12 \div 2 = 6$ cm **[B1]**

Calculates area of the hexagonal cross-section $= 6 \times \left(\frac{1}{2} \times 6 \times 6 \times \sin(60°)\right) = 54\sqrt{3}$ (or 93.53...) **[M1]**

Uses Volume = Area × Length to find Length $= \frac{450}{54\sqrt{3}}$ **[M1]**

4.81 (or 4.811...) **[A1]**

(ii) Method for total area = 2 × Base Area + 6 × Lateral Rectangular Area **[M1]**

Substitutes values: $2 \times 54\sqrt{3} + 6 \times (6 \times 4.811...)$ **[M1]**

Simplifies to $108\sqrt{3} + 100\sqrt{3}$ (or 187.06... + 173.2...) **[M1]**

360 (or 360.2... to 360.3...) **[A1]**

[Total Marks: 10]

21 **(a)** Name: kite **[B1]**

Order of rotational symmetry = 1 **[B1]**

(b) The line of symmetry AC is the perpendicular bisector of BD (or B is the reflection of D in AC) **[B1]**

(c) **(i)** Angle $BAM = \dfrac{106^\circ}{2} = 53^\circ$ **[B1]**

$BM = 9\sin 53^\circ \ (= 7.1877\ldots)$ **[M1]**

$BD = 2 \times 7.1877\cdots = 14.4$ **[A1]**

(ii) $\sin(\text{angle } BCM) = \dfrac{7.1877\ldots}{15} \ (= 0.47918\ldots)$ **[M1]**

Angle $BCM = 28.63^\circ$ **[A1]**

Angle $BCD = 2 \times 28.63^\circ = 57.3$ **[A1]**

(d) One plane along AC through the length of the prism and one plane bisecting the prism perpendicular to its length **[M1]**

2 **[A1]**

[Total Marks: 11]